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AI Training
Kuehne + Nagel

Setup Addendum

Workshop 3 · Giving the LLM Your Data (RAG)

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Workshop 3 · Giving the LLM Your Data (RAG) · 22 October 2026, 08:30-12:30

Workshop 3 builds a complete RAG pipeline on your laptop, with the same environment as Workshop 2 plus four additions: the Workshop 3 notebook folder, the embedding model **EmbeddingGemma 300M**, three Python packages, and a small reranking model (80 MB). Everything runs locally; nothing is sent outside the laptop during the session.

Estimated time: **10 minutes**, mostly downloads. **Do it before 08:30 on 22 October**, with an internet connection. If any step fails, send the error message to the trainer before the session.

Starting point: the Workshop 2 setup works: Ollama or LM Studio with Ministral 3 3B (or Gemma 4 E2B), and the Python environment of the Workshop 2 notebook.

1. The notebook folder

Copy the folder **K + N - W3 - Notebook** (sent with this addendum) next to your Workshop 2 notebook folder. It contains the notebook `kn_w3_rag.ipynb`, `helpers.py`, `check_embeddings.py` and the data in `data/`. Do not rename or move the files inside it.

2. EmbeddingGemma 300M

Skip this section if you already followed the Embedding Model Installation Guide.

Runtime	Command	Check
Ollama (0.11.10 or later)	`ollama pull embeddinggemma`	`ollama list` shows `embeddinggemma:latest`
LM Studio	`lms get google/embedding-gemma-300m`	`lms ls` lists it under the embedding models

Not EmbeddingGemma 2. LM Studio also offers *EmbeddingGemma 2*, a newer model. The training uses EmbeddingGemma 300M on every laptop so that everyone gets the same results. If you already have EmbeddingGemma 2, keep it and download EmbeddingGemma 300M as well. With LM Studio, the local server must be on (Developer tab, **Start server**, port 1234).

3. The three packages

Install them **into the same Python environment** you used for the Workshop 2 notebook:

Package	Used for
`langchain-core`	Documents, retrievers, vector store and chains (the frame of the notebook)
`langchain-text-splitters`	Cutting documents into chunks
`fastembed`	The reranker of unit 9

3.1 If your environment is a `.venv` made with Python (the Workshop 2 setup guide)

From your Workshop 2 notebook folder, the one that contains `.venv`:

```
# macOS / Linux
.venv/bin/python -m pip install langchain-core langchain-text-splitters fastembed

# Windows (PowerShell)
.venv\Scripts\python -m pip install langchain-core langchain-text-splitters fastembed
```

3.2 If your environment was made with uv

An environment made with `uv` has no `pip` of its own: the message is *No module named pip*. Install with uv instead, giving the path of the environment's Python:

```
uv pip install --python PATH-TO-THE-ENVIRONMENT/bin/python langchain-core langchain-text-splitters fastembed
```

On Windows the Python of the environment is `PATH-TO-THE-ENVIRONMENT\Scripts\python.exe`.

3.3 Not sure which Python the notebook uses?

Open any notebook with your Workshop 2 kernel and run this cell. It prints the path of the kernel's Python and installs the packages into exactly that one:

```
import sys, subprocess
print(sys.executable)
subprocess.run([sys.executable, "-m", "pip", "install", "langchain-core", "langchain-text-splitters", "fastembed"])
```

If it prints *No module named pip*, use section 3.2 with the path it printed.

4. Final check (2 minutes)

1. Start Ollama, or the LM Studio server.
2. Open `kn_w3_rag.ipynb` from the Workshop 3 folder, with your Workshop 2 kernel.
3. Run the cells of **section 0** in order, then restart the kernel once (Kernel → Restart) and run section 0 again.

The check cell must print **READY** and the pre-flight cell **PRE-FLIGHT OK: libraries and reranker ready**. The pre-flight downloads the reranking model (about 80 MB) the first time, so it needs the internet once. It is saved in the folder `models/` inside the Workshop 3 notebook folder: do not delete it, and the notebook then works offline.

- Section 0 prints **READY**
- Section 0 prints **PRE-FLIGHT OK**
- Turn off Wi-Fi, restart the kernel and run section 0 again: both still print. Turn Wi-Fi back on.

5. Common problems

Symptom	Likely cause	What to do
`No module named 'langchain_core'` after installing	The packages went into another Python, not the kernel's	Run the cell in section 3.3, then restart the kernel
`No module named pip`	The environment was made with uv	Install with `uv pip install --python …` (section 3.2)
`fastembed` fails to install	A Python version it has no build for	Send the error to the trainer; only unit 9 needs it, every other section works without it
The pre-flight cannot download the reranker	No internet, or a company proxy blocks the download	Run section 0 once from a network without the proxy; otherwise tell the trainer
`Embedding model: NOT FOUND`	EmbeddingGemma 300M is not installed, or the runtime is not running	Section 2; start Ollama or the LM Studio server
`EmbeddingGemma 2 is installed, but the training uses EmbeddingGemma 300M`	LM Studio downloaded the newer model	Download EmbeddingGemma 300M as well (section 2)
`Chat model: NOT FOUND`	The Workshop 2 model is not loaded or the runtime is off	Start the runtime; the chat model is needed from unit 7 on

Privacy

Everything in this addendum runs on your laptop. The downloads (the packages, EmbeddingGemma and the reranker) happen once, before the session. During the session the notebook only talks to your local Ollama or LM Studio, and the data in `data/` is fictitious.